

Robots in Context

Programming Exercise 5

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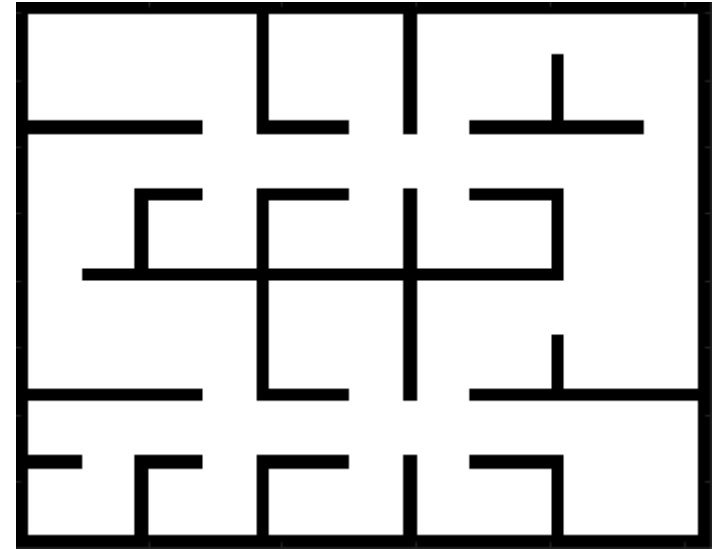
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A* search on a grid map

- Image
 - White represents free space, black represents occupied space
- You can use other colors to mark the paths



Programming Exercise - Task

- Implement a program which
 - generates a random start + goal configurations
 - generates a roadmap
 - implements an A*-search
 - reports the optimal path
- Optional
 - You may implement Dijkstra too and compare
 - You may choose different maps too, e.g.:
 - A) with lots of open space or B) a maze with few valid solutions



Programming Exercise 2 - Reporting

- General: 7ish minute presentation
 - Explain how you wanted to solve the problem; which bug algorithm(s), etc.
 - Implementation: Explain how you solved it, i.e. what tools (e.g. Matlab) you used
 - Evaluation: outline your evaluation strategy and show results. Consider potentially how to assess the performance and how to report it.
 - Again: Mention also challenges
 - Most likely someone else in this room will face the same problems
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